

```
!pip install bertopic
!pip install gensim
from google.colab import drive
drive.mount('/content/drive')
import sys
from bertopic import BERTopic
from umap import UMAP
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.cluster import KMeans
```

```
Requirement already satisfied: hdbscan==0.8.29 in /usr/local/lib/python3.12/dist-packages (from bertopic) (0.8.41)
Requirement already satisfied: umap-learn==0.5.0 in /usr/local/lib/python3.12/dist-packages (from bertopic) (0.5.11)
Requirement already satisfied: numpy==1.20.0 in /usr/local/lib/python3.12/dist-packages (from bertopic) (2.0.2)
Requirement already satisfied: pandas==1.5 in /usr/local/lib/python3.12/dist-packages (from bertopic) (2.2.2)
Requirement already satisfied: plotly==4.7.0 in /usr/local/lib/python3.12/dist-packages (from bertopic) (5.24.1)
Requirement already satisfied: scikit-learn==1.0 in /usr/local/lib/python3.12/dist-packages (from bertopic) (1.6.1)
Requirement already satisfied: sentence-transformers==0.4.1 in /usr/local/lib/python3.12/dist-packages (from bertopic) (0.4.1)
Requirement already satisfied: tqdm==4.41.1 in /usr/local/lib/python3.12/dist-packages (from bertopic) (4.67.3)
Requirement already satisfied: llvmlite>0.36.0 in /usr/local/lib/python3.12/dist-packages (from bertopic) (0.43.0)
Requirement already satisfied: scipy==1.0 in /usr/local/lib/python3.12/dist-packages (from hdbscan==0.8.29->bertopic) (1.10.1)
Requirement already satisfied: joblib==1.0 in /usr/local/lib/python3.12/dist-packages (from hdbscan==0.8.29->bertopic) (1.3.2)
Requirement already satisfied: python-dateutil==2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas==1.5->bertopic) (2.8.2)
Requirement already satisfied: pytz==2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas==1.5->bertopic) (2020.1)
Requirement already satisfied: tzdata==2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas==1.5->bertopic) (2022.7)
Requirement already satisfied: tenacity==6.2.0 in /usr/local/lib/python3.12/dist-packages (from plotly==4.7.0->bertopic) (6.2.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from plotly==4.7.0->bertopic) (24.1)
Requirement already satisfied: threadpoolctl==3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn==1.0->bertopic) (3.1.0)
Requirement already satisfied: transformers<6.0.0,>=4.41.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers==0.4.1->bertopic) (4.41.0)
Requirement already satisfied: huggingface-hub==0.20.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers==0.4.1->bertopic) (0.20.0)
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Requirement already satisfied: numba==0.51.2 in /usr/local/lib/python3.12/dist-packages (from umap-learn==0.5.0->bertopic) (0.51.2)
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Requirement already satisfied: fsspec==2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub==0.20.0->bertopic) (2023.5.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub==0.20.0->bertopic) (1.2.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub==0.20.0->bertopic) (0.23.0)
Requirement already satisfied: pyyaml==5.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub==0.20.0->bertopic) (5.1)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub==0.20.0->bertopic) (0.12.1)
Requirement already satisfied: six==1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil==2.8.2->pandas==1.5->bertopic) (1.5)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch==1.11.0->sentence-transformers==0.4.1->bertopic) (59.0.0)
Requirement already satisfied: sympy==1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch==1.11.0->sentence-transformers==0.4.1->bertopic) (1.13.3)
Requirement already satisfied: networkx==2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch==1.11.0->sentence-transformers==0.4.1->bertopic) (2.5.1)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch==1.11.0->sentence-transformers==0.4.1->bertopic) (3.1.2)
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Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub==0.20.0->bertopic) (1.0.5)
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Requirement already satisfied: h11==0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub==0.20.0->bertopic) (0.16)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy==1.13.3->torch==1.11.0->sentence-transformers==0.4.1->bertopic) (1.3.0)
Requirement already satisfied: MarkupSafe==2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->torch==1.11.0->sentence-transformers==0.4.1->bertopic) (2.0.1)
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Requirement already satisfied: rich==12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub==0.20.0->bertopic) (12.3.0)
Requirement already satisfied: annotated-doc==0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub==0.20.0->bertopic) (0.2)
Requirement already satisfied: markdown-it-py==2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich==12.3.0->typer->huggingface-hub==0.20.0->bertopic) (2.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich==12.3.0->typer->huggingface-hub==0.20.0->bertopic) (2.13.0)
Requirement already satisfied: mdurl==0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py==2.2.0->rich==12.3.0->typer->huggingface-hub==0.20.0->bertopic) (0.1)
Requirement already satisfied: gensim in /usr/local/lib/python3.12/dist-packages (4.4.0)
Requirement already satisfied: numpy==1.18.5 in /usr/local/lib/python3.12/dist-packages (from gensim) (2.0.2)
Requirement already satisfied: scipy==1.7.0 in /usr/local/lib/python3.12/dist-packages (from gensim) (1.16.3)
Requirement already satisfied: smart_open==1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim) (7.5.1)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open==1.8.1->gensim) (2.0.0)
Drive already mounted at /content/drive: to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

```
%run "/content/drive/MyDrive/Colab Notebooks/CombinedTrainingDataset.ipynb"
%run "/content/drive/MyDrive/Colab Notebooks/StopWords.ipynb"
```

```
Using Colab cache for faster access to the 'liberals-vs-conservatives-on-reddit-13000-posts' dataset.
Using Colab cache for faster access to the '1-million-reddit-comments-from-40-subreddits' dataset.
Successfully combined!
Dataset 1 rows: 12854
Dataset 2 ('politics' only) rows: 25000
Total combined rows: 37854
```

```
vectorizer_model = CountVectorizer(stop_words=my_stop_words, ngram_range=(1, 2), min_df=10, token_pattern=r'(?u)\b[a-
```

```

seed_topic_list = [
    # 1. Inflation & Cost of Living
    ["inflation", "prices", "cost", "economy", "fed", "recession", "gas", "groceries",
     "purchasing", "power", "interest", "rates", "supply", "chain", "deficit",
     "spending", "cpi", "consumer", "index", "expensive", "affordability"],
    # 2. Taxes & Wealth Distribution
    ["tax", "taxes", "irs", "wealth", "billionaires", "loophole", "income", "corporate",
     "brackets", "capital", "gains", "revenue", "billionaire", "millionaires", "fair",
     "share", "audit", "deduction", "estates", "rich", "taxation"],
    # 3. Labor Rights & Unions
    ["labor", "unions", "strike", "workers", "wages", "minimum", "pay", "benefits",
     "bargaining", "unionization", "guild", "workplace", "employment", "overtime",
     "safety", "pension", "salaries", "staffing", "contract", "exploitation"],
    # 4. Housing & Real Estate
    ["housing", "rent", "mortgage", "zoning", "homelessness", "affordable", "eviction",
     "landlord", "tenants", "property", "real", "estate", "development", "market",
     "shortage", "equity", "gentrification", "shelter", "units", "inventory"],
    # 5. Student Debt & Education
    ["debt", "student", "loans", "forgiveness", "tuition", "college", "university",
     "degree", "graduates", "interest", "borrowers", "education", "academic",
     "scholarship", "fafsa", "repayment", "cancel", "default", "higher", "learning"],
    # 6. Healthcare & Insurance
    ["insurance", "healthcare", "medicare", "hospital", "coverage", "premiums", "prescription",
     "medicaid", "deductible", "copay", "provider", "doctor", "physician", "clinic", "treatment",
     "patient", "medical", "pharma", "aca", "obamacare"],
    # 7. Abortion & Reproductive Rights
    ["abortion", "roe", "wade", "planned", "parenthood", "fetus", "reproductive", "pregnancy",
     "choice", "prolife", "prochoice", "clinic", "viability", "contraception", "birth",
     "autonomy", "rights", "women", "dobbs", "legal"],
    # 8. Pandemic & Public Health
    ["covid", "pandemic", "vaccine", "mandate", "mask", "fauci", "virus", "health",
     "quarantine", "lockdown", "outbreak", "moderna", "pfizer", "booster", "cdc",
     "transmission", "distancing", "public", "emergency", "variants"],
    # 9. Drug Policy & Decriminalization
    ["drugs", "legalization", "marijuana", "weed", "decriminalization", "addiction", "overdose",
     "cannabis", "opioid", "fentanyl", "narcotics", "treatment", "dispensary", "rehab",
     "recreational", "controlled", "substances", "prison", "sentencing", "possession"],
    # 10. Gun Control & Second Amendment
    ["gun", "guns", "nra", "amendment", "firearm", "background", "checks", "shooting",
     "violence", "weapons", "rifle", "pistol", "ban", "arms", "safety",
     "massacre", "shooter", "regulation", "carry", "concealed"],
    # 11. LGBTQ+ & Gender Issues
    ["lgbtq", "trans", "transgender", "gay", "marriage", "gender", "rights", "pride",
     "queer", "homosexuality", "bisexual", "pronouns", "identity", "nonbinary", "transition",
     "equality", "discrimination", "hormone", "bathroom", "ban"],
    # 12. Racial Justice & Civil Rights
    ["race", "racism", "systemic", "blm", "black", "white", "diversity", "equity",
     "inclusion", "prejudice", "minority", "supremacy", "segregation", "reparations", "privilege",
     "civil", "rights", "marginalized", "profiling", "justice"],
    # 13. Policing & Law Enforcement
    ["police", "cops", "brutality", "reform", "defund", "officer", "arrest", "accountability",
     "sheriff", "patrol", "shooting", "force", "misconduct", "qualified", "immunity",
     "detective", "badge", "law", "enforcement", "swat"],
    # 14. Criminal Justice & Prison Reform
    ["prison", "justice", "criminal", "reform", "incarceration", "sentence", "inmates",
     "jail", "parole", "probation", "felony", "misdemeanor", "conviction", "recidivism", "rehab",
     "solitary", "warden", "exonerated", "private", "prisons"],
    # 15. Climate Change & Energy
    ["climate", "warming", "emissions", "carbon", "fossil", "green", "energy", "oil",
     "renewable", "solar", "wind", "methane", "epa", "environment", "pollution",
     "sustainability", "gas", "coal", "grid", "electric"],
    # 16. Infrastructure & Transportation
    ["infrastructure", "roads", "bridges", "transit", "rail", "amtrak", "grid", "transportation",
     "highway", "subway", "trains", "tunnel", "construction", "pipes", "broadband",
     "water", "sewage", "logistics", "electric", "vehicles"],
    # 17. Voting Rights & Elections
    ["voting", "voters", "election", "fraud", "ballot", "gerrymandering", "suppression",
     "electoral", "college", "registration", "absentee", "primary", "midterms", "polling", "census",
     "districts", "integrity", "disenfranchisement", "turnout", "early"],
    # 18. Supreme Court & Judiciary
    ["supreme", "court", "scotus", "justices", "judge", "ruling", "bench", "nominee",
     "jurisdiction", "unconstitutional", "precedent", "confirmation", "docket", "circuit", "legal",
     "opinion", "constitution", "judicial", "overturn", "appeal"],
    # 19. Campaign Finance & Corruption
    ["campaign", "finance", "citizens", "united", "pac", "lobbying", "donors", "corruption",

```

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"superpac", "bribery", "ethics", "lobbyist", "funding", "special", "interests",
"dark", "money", "disclosure", "clout", "quid"],
# 20. Executive Power & Investigations
["impeachment", "subpoena", "treason", "investigation", "committee", "hearing", "testimony",
"executive", "privilege", "pardon", "cabinet", "veto", "prosecutor", "contempt", "oversight",
"scandal", "witness", "perjury", "doj", "order"],
# 21. Immigration & Border Security
["border", "immigration", "immigrants", "wall", "asylum", "ice", "deportation", "migrants",
"citizenship", "visa", "illegal", "legal", "daca", "patrol", "crossing",
"refugee", "customs", "undocumented", "naturalization", "h1b"],
# 22. Russia & Ukraine Conflict
["ukraine", "russia", "putin", "zelensky", "nato", "war", "sanctions", "invasion",
"kremlin", "kyiv", "moscow", "missiles", "intelligence", "offensive", "tanks",
"aid", "weapons", "ceasefire", "territory", "victory"],
# 23. China & Global Trade
["china", "taiwan", "tariffs", "ccp", "trade", "beijing", "espionage", "manufacturing",
"semiconductors", "supply", "chain", "exports", "imports", "economy", "sanctions",
"tensions", "dominance", "competition", "shipping", "xi"],
# 24. Middle East Relations
["israel", "palestine", "gaza", "hamas", "middle", "east", "jewish", "antisemitism",
"conflict", "iran", "jerusalem", "tel", "aviv", "settlements", "apartheid",
"hezbollah", "oil", "syria", "treaty", "hostages"],
# 25. Big Tech, Privacy & Social Media
["tech", "privacy", "tiktok", "facebook", "censorship", "algorithm", "data", "monopoly",
"silicon", "valley", "encryption", "surveillance", "platform", "twitter", "google",
"antitrust", "regulation", "section", "230", "internet"]
]

```

```

docs = combined_data['Combined_Content'].tolist()

umap_model = UMAP( n_neighbors=15, n_components=5, min_dist=0.0, metric='cosine', random_state=42)

cluster_model = KMeans(n_clusters=49, random_state=42)

topic_model = BERTopic(hdbscan_model=cluster_model, umap_model=umap_model, vectorizer_model=vectorizer_model, seed_topics,
probs = topic_model.fit_transform(docs)
freq = topic_model.get_topic_info()

```

2026-03-10 23:53:07,383 - BERTopic - Embedding - Transforming documents to embeddings.

Loading weights: 100%

103/103 [00:00<00:00, 197.46it/s, Materializing param=pooler.dense.weight]

BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2

Key	Status	
embeddings.position_ids	UNEXPECTED	

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.

Batches: 100%

1183/1183 [18:11<00:00, 9.51it/s]

2026-03-11 00:11:24,550 - BERTopic - Embedding - Completed ✓

2026-03-11 00:11:24,552 - BERTopic - Guided - Find embeddings highly related to seeded topics.

Batches: 100%

1/1 [00:00<00:00, 2.59it/s]

2026-03-11 00:11:25,274 - BERTopic - Guided - Completed ✓

2026-03-11 00:11:25,275 - BERTopic - Dimensionality - Fitting the dimensionality reduction algorithm

2026-03-11 00:12:40,699 - BERTopic - Dimensionality - Completed ✓

2026-03-11 00:12:40,701 - BERTopic - Cluster - Start clustering the reduced embeddings

2026-03-11 00:12:41,067 - BERTopic - Cluster - Completed ✓

2026-03-11 00:12:41,080 - BERTopic - Representation - Fine-tuning topics using representation models.

2026-03-11 00:12:44,483 - BERTopic - Representation - Completed ✓

```

topic_info = topic_model.get_topic_info()
formatted_labels = []

for index, row in topic_info.iterrows():
    topic_id = row['Topic']
    words = row['Representation']

    if topic_id == -1:
        clean_label = "Other / Irrelevant / Noise"
    else:
        clean_label = ", ".join(words)
    formatted_labels.append(clean_label)

for i, label in enumerate(formatted_labels):

```

```
print(f"Index {i}: {label}")
```

```
llm_topic_list = formatted_labels
```

```
Index 0: talking, theyre, exactly, real, funny, mean, pretty, god, okay, way
Index 1: senate, house, mcconnell, reagan, theyre, power, white, law, democratic, run
Index 2: capitalism, socialism, social, socialist, china, communist, market, communism, democratic, democracy
Index 3: lie, congress, lying, resign, lies, returns, fraud, oath, jail, hearing
Index 4: ted, lieu, hearing, water, buttery, movie, way, males, dick, dude
Index 5: ukraine, russia, putin, russian, war, nato, invasion, sanctions, military, conflict
Index 6: mueller, barr, letter, barrs, report, summary, investigation, special, muellers, public
Index 7: mueller, letter, public, doj, report, investigation, department, muellers, released, mueller report
Index 8: candidate, primary, warren, poll, candidates, voters, nominee, progressive, polls, pete
Index 9: doj, perjury, congress, testify, contempt, indicted, justice, indict, sitting, mueller
Index 10: report, summary, context, title, headline, word, lie, pages, applied, redacted
Index 11: democracy, theyre, god, education, history, libs, far, shapiro, way, fear
Index 12: voters, electoral, elections, primary, ballot, polls, poll, democratic, democracy, rights
Index 13: barr, barrs, credibility, testimony, congress, lied, william barr, lying, william, impeached
Index 14: obstruction, collusion, investigation, report, evidence, justice, crime, campaign, conspiracy, russian
Index 15: nazis, israel, fascism, nazi, leftist, fascist, social, war, jewish, freedom
Index 16: pelosi, warren, impeachment, buttigieg, candidate, nancy, campaign, pete, progressive, win
Index 17: workers, labor, unions, strike, wage, union, minimum, wages, pay, work
Index 18: graham, lindsey, lindsey graham, emails, lindsay, talking, hillarys, buttery, males, clintons
Index 19: black, white, race, racism, racist, rights, slavery, police, minorities, history
Index 20: economy, inflation, market, economic, money, growth, price, income, labor, prices
Index 21: feminist, gender, sex, female, rape, trans, male, lgbtq, sexual, feminists
Index 22: news, fox, fox news, media, fake, fake news, propaganda, cnn, conspiracy, msnbc
Index 23: climate, border, canadian, canada, immigration, climate change, oil, energy, wall, convoy
Index 24: impeachment, impeach, senate, impeached, nixon, house, convict, proceedings, office, trial
Index 25: account, content, approved, automatically, internet, organizations, contain, form, action, site
Index 26: tax, taxes, wealth, rich, income, money, pay, billionaires, wealthy, gains
Index 27: money, corruption, funding, donors, campaign, donation, corporations, donations, corporate, subsidies
Index 28: insurance, healthcare, health, medicare, pay, care, health insurance, health care, medical, doctor
Index 29: abortion, baby, birth, womens, rights, child, texas, babies, children, doctors
Index 30: venezuela, coup, war, cuba, military, oil, opposition, korea, sanctions, cia
Index 31: covid, vaccine, pandemic, medical, health, mask, masks, public health, mandate, mandates
Index 32: prison, prisoners, jail, rights, prisons, law, crime, sentence, criminal, justice
Index 33: gun, guns, amendment, violence, shooting, armed, rights, mass, control, law
Index 34: police, officer, cops, shooting, officers, enforcement, safety, law, arrest, law enforcement
Index 35: court, supreme, supreme court, judge, judges, courts, prosecutors, justice, judicial, case
Index 36: land, housing, property, rent, tax, homeless, estate, market, capital, ownership
Index 37: cpac, florida, alabama, school, teachers, recall, schools, teacher, governor, california
Index 38: ban, civil, wishing, insults, shill, hate speech, advocating, violation, harm, permanent
Index 39: student, debt, college, education, loans, tuition, loan, pay, cancel, school
Index 40: marijuana, drugs, drug, weed, mental, cancer, pot, health, mental health, jail
Index 41: thank, length, updates, temporarily, news, concerning, removing, discussed, scale, participating
Index 42: submitted, mistakes, participating, following, message, removed, thank, reason, question, questions
Index 43: ban, civil, wishing, insults, shill, hate speech, advocating, violation, harm, permanent
Index 44: ban, civil, wishing, insults, shill, hate speech, advocating, violation, harm, permanent
Index 45: removed, questions, explicitly, participating, reasons, following, message, current, thank, question
Index 46: ban, civil, wishing, insults, shill, hate speech, advocating, violation, harm, permanent
Index 47: ban, civil, wishing, insults, shill, hate speech, advocating, violation, harm, permanent
Index 48: forums, , , , , , , ,
```

```
info = topic_model.get_topic_info()
print(info)
```

```

37 [cpac, florida, alabama, school, teachers, rec...
38 [ban, civil, wishing, insults, skill, hate spe...
39 [student, debt, college, education, loans, tui...
40 [marijuana, drugs, drug, weed, mental, cancer,...
41 [thank, length, updates, temporarily, news, co...
42 [submitted, mistakes, participating, following...
43 [ban, civil, wishing, insults, skill, hate spe...
44 [ban, civil, wishing, insults, skill, hate spe...
45 [removed, questions, explicitly, participating...
46 [ban, civil, wishing, insults, skill, hate spe...
47 [ban, civil, wishing, insults, skill, hate spe...
48 [forums, , , , , , , , ]

```

Representative_Docs

```

0 [This is the only talking point you mad hatter...
1 [Senate today, House tomorrow., Also, everyone...
2 [What would change in the past and capitalism ...
3 [Well, it's a crime to lie to Congress, plus u...
4 [Ted Lieu is the soundbyte king., I fee like T...
5 [Tulsi Gabbard Says Biden Administration Wants...
6 [Did *anyone* read the actual report on what B...
7 [&gt; "The summary letter the Department sent ...
8 [The Bernie/Warren hardcore supporters are goi...
9 [Serious question, how can the DoJ hold it up...
10 [Hi `TYLER_WAS_ROBBED`. Thank you for particip...
11 [All cultures of anti-intellectualism such as ...
12 [Terry McAuliffe wins democratic Virginia prim...
13 [Misinterpretations because of Barr's omission...
14 [Mueller should have just called it obstructio...
15 [Fascism needs help from moderates. Fascism ca...
16 [Leave Warren out of this., Wow and people say...

```

```

topics_to_merge = [
    [38, 43, 44, 46, 47 48],
    [6,7,13],
    [8, 16],
    [25, 42, 42, 45]
]
topic_model.merge_topics(docs, topics_to_merge)

```

```

formatted_labels = []

for topic_id in topic_model.get_topics():

    words_with_scores = topic_model.get_topic(topic_id)
    words_only = [word for word, score in words_with_scores[:20]]
    if topic_id == -1:
        clean_label = "Other / Irrelevant / Noise"
    else:
        clean_label = ", ".join(words_only)
    formatted_labels.append(clean_label)
for i, label in enumerate(formatted_labels):
    print(f"Index {i}: {label}")

```

```

Index 0: mueller, barr, report, letter, investigation, doj, congress, public, summary, special
Index 1: voters, candidate, primary, warren, candidates, poll, win, polls, democratic, campaign
Index 2: removed, participating, thank, questions, message, following, question, mistakes, reasons, reason
Index 3: senate, house, mcconnell, reagan, theyre, power, white, law, democratic, way
Index 4: capitalism, socialism, social, socialist, china, market, communist, democratic, communism, democracy
Index 5: congress, lie, lying, resign, lies, returns, fraud, oath, jail, hearing
Index 6: ukraine, russia, putin, russian, war, nato, invasion, sanctions, military, conflict
Index 7: democracy, theyre, god, education, history, far, way, shapiro, different, war
Index 8: nazis, israel, fascism, nazi, leftist, fascist, social, war, jewish, freedom
Index 9: workers, labor, unions, strike, wage, union, minimum, wages, pay, work
Index 10: black, white, race, racism, racist, rights, slavery, police, history, minorities
Index 11: economy, inflation, market, money, economic, growth, price, income, labor, value
Index 12: feminist, gender, sex, female, rape, male, trans, sexual, lgbtq, radical
Index 13: news, fox, fox news, media, fake, fake news, propaganda, cnn, conspiracy, msnbc
Index 14: climate, border, canada, canadian, immigration, climate change, oil, energy, wall, protests
Index 15: impeachment, senate, nixon, house, office, trial, congress, criminal, crimes, justice
Index 16: account, content, approved, automatically, internet, organizations, contain, form, action, social
Index 17: tax, taxes, wealth, rich, income, money, pay, billionaires, wealthy, gains
Index 18: money, corruption, funding, donors, campaign, corporations, donations, corporate, subsidies, farmers
Index 19: insurance, healthcare, health, medicare, pay, care, health insurance, health care, medical, doctor
Index 20: abortion, baby, birth, rights, womens, child, texas, children, babies, law
Index 21: venezuela, coup, war, cuba, military, oil, opposition, sanctions, korea, cia
Index 22: ban, civil, skill, advocating, violation, harm, permanent, users, contact, automatically
Index 23: covid, vaccine, pandemic, medical, health, mask, masks, public health, mandate, mandates
Index 24: prison, prisoners, jail, rights, law, prisons, crime, sentence, criminal, justice
Index 25: gun, guns, amendment, violence, shooting, rights, armed, mass, law, control
Index 26: police, officer, cops, shooting, officers, safety, enforcement, law, arrest, law enforcement

```

Index 27: court, supreme, supreme court, judge, judges, courts, justice, judicial, case, resign
 Index 28: land, housing, property, rent, tax, homeless, estate, market, capital, ownership
 Index 29: cpac, florida, alabama, school, teachers, recall, schools, teacher, governor, california
 Index 30: student, debt, college, education, loans, tuition, loan, pay, cancel, school
 Index 31: marijuana, drugs, drug, mental, cancer, pot, health, jail, mental health, legal
 Index 32: thank, news, updates, temporarily, concerning, removing, discussed, participating, scale, hello

```
topic_mapping = {
    0: "INVESTIGATION: Government Investigations",
    1: "ELECTIONS: Campaigns & Primaries",
    2: "MODERATION: Community Interaction Messages",
    3: "POLITICS: Legislative Branch & Governance",
    4: "IDEOLOGY: Economic & Political Systems",
    5: "POLITICS: Ethical Misconduct & Accountability",
    6: "FOREIGN POLICY: Russia-Ukraine Conflict",
    7: "CULTURE: Values & Historical Identity",
    8: "IDEOLOGY: Extremism & Global Conflict",
    9: "LABOR: Workers Rights & Unions",
    10: "SOCIAL: Race & Civil Rights",
    11: "ECONOMY: Economics & Markets",
    12: "SOCIAL: Gender & Identity",
    13: "MEDIA: News Outlets & Propaganda",
    14: "ENVIRONMENT & IMMIGRATION: Climate & Borders",
    15: "POLITICS: Impeachment & Presidential Trials",
    16: "TECH: Social Media & Platform Rules",
    17: "ECONOMY: Wealth & Taxation",
    18: "POLITICS: Campaign Finance & Corruption",
    19: "HEALTH: Healthcare Policy & Insurance",
    20: "SOCIAL: Reproductive Rights & Abortion",
    21: "FOREIGN POLICY: Latin America & Global Policy",
    22: "POLICY: Violations & Bans",
    23: "HEALTH: COVID-19 & Public Health",
    24: "JUSTICE: Criminal Justice & Incarceration",
    25: "JUSTICE: Gun Control & 2nd Amendment",
    26: "JUSTICE: Policing & Law Enforcement",
    27: "JUSTICE: Supreme Court & Legal Rulings",
    28: "ECONOMY: Housing & Real Estate",
    29: "POLITICS: State Governance & Education",
    30: "SOCIAL: Student Debt & Higher Education",
    31: "JUSTICE: Drug Policy & Legalization",
    32: "NOISE: General Conversational Filler"
}
```

```
import pickle
save_path = '/content/drive/MyDrive/K-means_bertopic_model_data.pkl'
import pickle
with open(save_path, 'wb') as f:
    pickle.dump({
        'vectorizer': vectorizer_model,
        'topic_model': topic_model,
        'human_labels': topic_mapping
    }, f)

print("BERTopic model and labels saved as 'K-means_bertopic_model_data.pkl'")
```

BERTopic model and labels saved as 'K-means_bertopic_model_data.pkl'

